

Sam Shelly

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AI Engineer focused on locally-hosted, secure AI systems for privacy-sensitive environments

EDUCATION

Vassar College

Aug 2022 – May 2026

Bachelor of Arts in Computer Science, Minor in Astronomy

Poughkeepsie, NY

GPA: 3.64/4.00 | Major GPA: 3.68/4.00 | *Coursework:* Data Structures & Algorithms, Analysis of Algorithms, Software Design, Compilers, Theory of Computation, Computer Organization

EXPERIENCE

AI Development Intern

May 2025 – Jan 2026

Innovative Computing Systems

Los Angeles, CA

- Hired directly by the COO to lead a Microsoft Copilot Studio Knowledge Agent pilot at a ~60-person company; worked full-time (30 hrs/week) as the primary builder on a 4-person team, meeting weekly with the COO to scope use cases.
- Built Python-based ingestion pipelines to index SharePoint content – including SOPs, the employee handbook, and company org hierarchy – for retrieval, and implemented an API integration connecting the agent to the company's work-ticket system, letting engineers instantly query prior work history on a given client 30% faster lookups.
- Rolled the agent out to a daily-use focus group of 15 users and began scoping it for a second prospective external client as part of early efforts to commercialize the tool.

Computer Science Dept. Coach

Jan 2024 – May 2026

Vassar College

Poughkeepsie, NY

- Selected for a formal, paid CS Dept Coach position (8 hrs/week), retained across 4 consecutive semesters spanning two course iterations under two different instructors.
- Led 3 weekly sessions on algorithms, data structures, and debugging in Java for up to 50 students per semester (200+ total), adapting pacing and session plans for struggling students.

PROJECTS

AI Knowledge Base RAG

GitHub

Python, FastAPI, LangChain, Qdrant, Ollama, Mistral 7B, BAAI/bge-large-en-v1.5, PyMuPDF, Docker, Streamlit, pytest, GitHub Actions

Closed-loop RAG system built entirely on locally-hosted, open-weight components (Mistral 7B via Ollama, Qdrant) with no external API calls – suited to air-gapped and other network-restricted deployments. Applied to query government/defense AI policy documents (NIST AI RMF, OMB M-25-21, DoD AI Strategy, GSA AI compliance materials), eliminating cloud dependency and controlling data exposure for secure, privacy-sensitive environments.

- By building a custom LLM-as-judge evaluation pipeline, achieved 0.95 retrieval recall and 0.92 answer similarity across 10 ground-truth QA pairs, visualized in a Streamlit dashboard.
- Ingests PDFs via PyMuPDF with overlapping chunking; embeds via BAAI/bge-large-en-v1.5 into Qdrant.
- Retrieves top-k chunks and generates grounded, cited answers via Mistral 7B (Ollama).
- Served via FastAPI REST API, containerized with Docker Compose, with CI/CD via GitHub Actions.

DataFest 2024 — Best Overall Analysis

4-person team, 48-hr competition

Python, pandas

Analyzed student-interaction data (video views, quiz attempts, textbook page views; $n \approx 1,857$ students) from an online statistics textbook to advise an ed-tech client on improving learning outcomes.

- Team built a multiple regression model ($R^2=0.47$) identifying quiz effort and video engagement as positive predictors of student performance, and quantified that textbook-only engagement and frequent answer-changing predicted lower scores.
- A teammate built a pandas-based directed dependency graph mapping how students navigated between textbook chapters/sections while studying, informing a recommendation to prepend review material to high-traffic “capstone” chapters.
- Authored the team's final write-up and slide presentation synthesizing both analyses into client-facing recommendations; won Best Overall Analysis over 20 competing teams from 3 schools.

PUBLICATIONS

Feedback-Driven Magnetic Field Evolution in MW-type Simulated Circumgalactic Medium — KNAC Undergraduate Symposium, 2024 | **Efficacy of NER Models on Fiction and Nonfiction Domains** — CMU 366 NLP, Vassar College, 2025

SKILLS

Languages: Python, Java, C, C++, C#, OCaml | **ML/AI:** PyTorch, HuggingFace Transformers, LangChain, Ollama, Qdrant | **Tools:** Docker, FastAPI, Streamlit, GitHub Actions, PyMuPDF, RISC-V, pytest